



University  
of Manitoba

# **Graphs – Introduction (theory) – 4**

**MATH 2740 – Mathematics of Data Science – Lecture 17**

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The University of Manitoba campuses are located on original lands of Anishinaabeg, Ininew, Anisininew, Dakota and Dene peoples, and on the National Homeland of the Red River Métis. We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

# Outline

## Trees



**Trees**



# Trees

## Definition 154 (Forest, trees and branches)

- ▶ A connected graph with no cycle is a **tree**
- ▶ A tree is a connected acyclic graph, its edges are called **branches**
- ▶ A graph (connected or not) without any cycle is a **forest**. Each component is a tree

(A forest is a graph whose connected components are trees)

# Is the “Karate graph” a tree?

```
is_acyclic(G_Z)
```

```
## [1] FALSE
```

```
is_tree(G_Z)
```

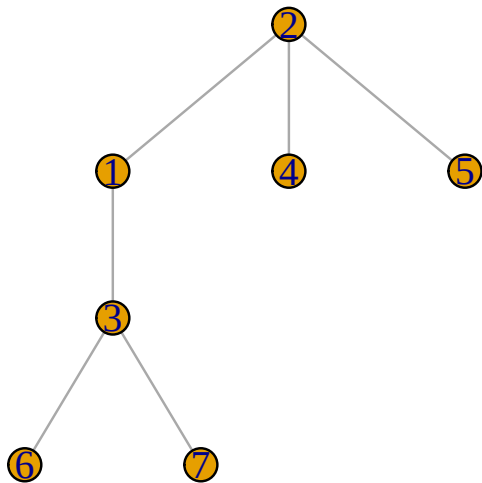
```
## [1] FALSE
```

So we need a friend to play with!

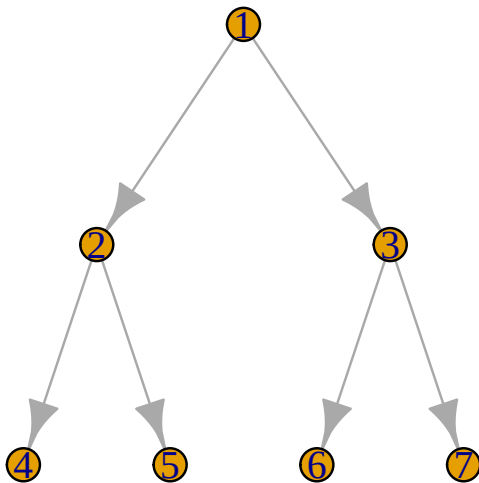
```
G_tu <- make_tree(7, 2, mode = "undirected")
```

```
G_td <- make_tree(7, 2)
```

## An undirected tree



## An out directed tree



## Property 155

- ▶ *Every edge of a tree is a bridge*
- ▶ *Given two vertices  $u$  and  $v$  of a tree, there is an unique path linking  $u$  to  $v$*
- ▶ *A tree with  $p$  vertices and  $q$  edges satisfies  $q = p - 1$ . Thus, a tree is minimally connected*

(First property: the deletion of any edge of a tree disconnects it)

# Every edge of a tree is a bridge

```
E(G_tu)

## + 6/6 edges from d9e8953:
## [1] 1--2 1--3 2--4 2--5 3--6 3--7

bridges(G_tu)

## + 6/6 edges from d9e8953:
## [1] 2--4 2--5 1--2 3--6 3--7 1--3

all(sort(E(G_tu)) == sort(bridges(G_tu)))

## [1] TRUE
```



# Spanning tree

## Definition 156 (Spanning tree)

A **spanning tree** of a connected graph  $G$  is a subgraph of  $G$  that contains all the vertices of  $G$  and is a tree.

A graph may have many spanning trees

# Minimal spanning tree

## Definition 157 (Value of a spanning tree)

The **value of a spanning tree**  $T$  of order  $p$  is

$$\sum_{i=1}^{p-1} f(e_i)$$

where  $f$  is the function that maps the edge set into  $\mathbb{R}$

## Definition 158 (Minimal spanning tree)

Let  $G$  be an undirected network, and let  $T$  be a **minimal spanning tree** of  $G$ . Then  $T$  is a spanning tree whose the value is minimum

## Algorithm to find a minimal spanning tree

Let  $G = (V(G), E(G))$  be an undirected network and  $T$  be a minimal spanning tree

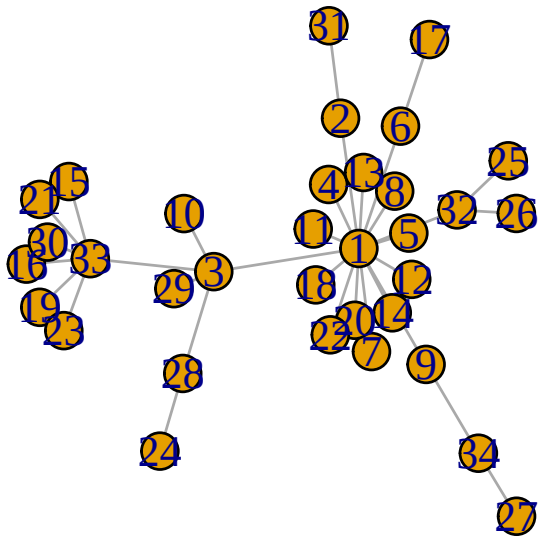
1. Sort the edges of  $G$  in increasing order by value
2.  $T = (V(G), \emptyset)$
3. For each edge  $e$  in sorted order if the endpoints of  $e$  are disconnected in  $T$  add  $e$  to  $T$

## Finding a minimal spanning tree of the Karate graph

The function `mst` finds minimal spanning trees, using distances if no edge weights are provided

```
G_mst = mst(G_Z)
```

# A minimal spanning tree of the Karate graph



# Minimal connector problem

- ▶ Model: a graph  $G$  such that edges represent all possible connections, and each edge has a positive value which represents its cost; an undirected network  $G$
- ▶ Solution: a minimal spanning tree  $T$  of  $G$ 
  - ▶ a spanning tree of  $G$  is a subgraph of  $G$  that contains all the vertices of  $G$  and is a tree.
  - ▶ the cost of the spanning tree is the sum of values of the edges of  $T$
  - ▶ a spanning tree such that no other spanning tree has a smaller cost is a minimal spanning tree.

## Theorem 159 (Characterisation of trees)

*$H = (V, U)$  a graph of order  $|V| = n > 2$ . The following are equivalent and all characterise a tree :*

- 1.  $H$  connected and has no cycles*
- 2.  $H$  has  $n - 1$  arcs and no cycles*
- 3.  $H$  connected and has exactly  $n - 1$  arcs*
- 4.  $H$  has no cycles, and if an arc is added to  $H$ , exactly one cycle is created*
- 5.  $H$  connected, and if any arc is removed, the remaining graph is not connected*
- 6. Every pair of vertices of  $H$  is connected by one and only one chain*

### Definition 160 (Pendant vertex)

A vertex is **pendant** if it is adjacent to exactly one other vertex

### Theorem 161

*A tree of order  $n \geq 2$  has at least two pendant vertices*



## Theorem 162

*A graph  $G = (V, U)$  has a partial graph that is a tree  $\iff G$  connected*

(A partial graph is a graph generated by a subset of the arcs)

# Spanning tree

Can build a spanning tree as follows:

- ▶ Consider any arc  $u_0$
- ▶ Find arc  $u_1$  that does not form a cycle with  $u_0$
- ▶ Find arc  $u_2$  that does not form a cycle with  $\{u_0, u_1\}$
- ▶ Continue
- ▶ When you cannot continue anymore, you have a spanning tree

### Definition 163 (Minimally connected graph)

$G$  is **minimally connected** if it is strongly connected and removal of any arc destroys strong-connectedness

A minimally connected graph is 1-graph without loops

### Definition 164 (Contraction)

$G = (V, U)$ . The **contraction** of the set  $A \subset V$  of vertices consists in replacing  $A$  by a single vertex  $a$  and replacing each arc into (resp. out of)  $A$  by an arc with same index into (resp. out of)  $a$

## Theorem 165

*$G$  minimally connected,  $A \subset V$  generating a strongly connected subgraph of  $G$ .  
Then the contraction of  $A$  gives a minimally connected graph*

# Arborescences

## Definition 166 (Root)

Vertex  $a \in V$  in  $G = (V, U)$  is a **root** if all vertices of  $G$  can be reached by paths *starting* from  $a$

Not all graphs have roots

## Definition 167 (Quasi-strong connectedness)

$G$  is **quasi-strongly connected** if  $\forall x, y \in V$ , exists  $z \in V$  (denoted  $z(x, y)$  to emphasize dependence on  $x, y$ ) from which there is a path to  $x$  and a path to  $y$

Strongly connected  $\implies$  quasi-strongly connected (take  $z(x, y) = x$ ); converse not true

Quasi-strongly connected  $\implies$  connected

# Arborescence

## Definition 168 (Arborescence)

An **arborescence** is a tree that has a root

## Lemma 169

$G = (V, U)$  has a root  $\iff G$  quasi-strongly connected

## Theorem 170

*H graph of order  $n > 1$ . TFAE (and all characterise an arborescence)*

- 1. H quasi-strongly connected without cycles*
- 2. H quasi-strongly connected with  $n - 1$  arcs*
- 3. H tree having a root a*
- 4.  $\exists a \in V$  s.t. all other vertices are connected with a by 1 and only 1 path from a*
- 5. H quasi-strongly connected and loses quasi-strong connectedness if any arc is removed*
- 6. H quasi-strongly connected and  $\exists a \in V$  s.t.*

$$d_H^-(a) = 0 \quad \text{and} \quad d_H^-(x) = 1, \quad \forall x \neq a$$

- 7. H has no cycles and  $\exists a \in V$  s.t.*

$$d_H^-(a) = 0 \quad \text{and} \quad d_H^-(x) = 1, \quad \forall x \neq a$$

### Theorem 171

*$G$  has a partial graph that is an arborescence  $\iff G$  quasi-strongly connected*

### Theorem 172

*$G = (V, E)$  simple connected graph and  $x_1 \in V$ . It is possible to direct all edges of  $E$  so that the resulting graph  $G_0 = (V, U)$  has a spanning tree  $H$  s.t.*

- 1.  $H$  is an arborescence with root  $x_1$*
- 2. The cycles associated with  $H$  are circuits*
- 3. The only elementary circuits of  $G_0$  are the cycles associated with  $H$*



# Counting trees

## Proposition 173

*$X$  a set with  $n$  distinct objects,  $n_1, \dots, n_p$  nonnegative integers s.t.  $n_1 + \dots + n_p = n$ . The number of ways to place the  $n$  objects into  $p$  boxes  $X_1, \dots, X_p$  containing  $n_1, \dots, n_p$  objects respectively is*

$$\binom{n}{n_1, \dots, n_p} = \frac{n!}{n_1! \cdots n_p!}$$

## Proposition 174 (Multinomial formula)

*Let  $a_1, \dots, a_p \in \mathbb{R}$  be  $p$  real numbers, then*

$$(a_1 + \dots + a_p)^n = \sum_{n_1, \dots, n_p \geq 0} \binom{n}{n_1, \dots, n_p} (a_1)^{n_1} \cdots (a_p)^{n_p}$$

## Theorem 175

Denote  $T(n; d_1, \dots, d_n)$  the number of distinct trees  $H$  with vertices  $x_1, \dots, x_n$  and with degrees  $d_H(x_1) = d_1, \dots, d_H(x_n) = d_n$ . Then

$$T(n; d_1, \dots, d_n) = \binom{n-2}{d_1-1, \dots, d_n-1}$$

## Theorem 176

The number of different trees with vertices  $x_1, \dots, x_n$  is  $n^{n-2}$

There is a whole industry of similar results (as well as for arborescences), but we will stop here. The main point is that we are talking about a large number of possibilities..